


Problem 1

$$X \sim \text{Gamma}(2, \lambda)$$

$$Y | X = x \sim \text{Uniform}[0, x]$$

$$\underline{f(x; 2, \lambda)} = \frac{\lambda^2 x e^{-\lambda x}}{\Gamma(2)} I_{[0, \infty)}(x)$$

$$f(y|x) = \frac{1}{x} I_{[0, x]}(y)$$

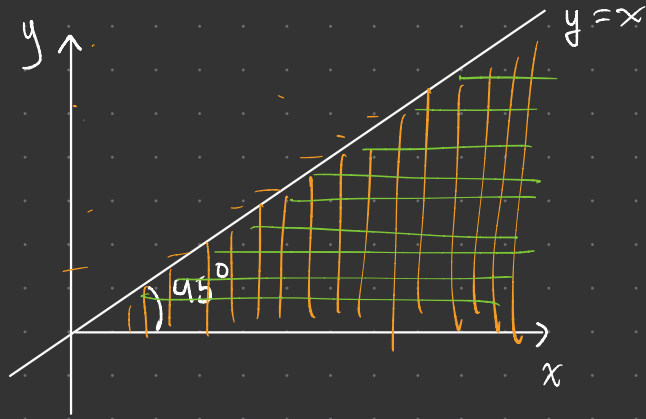
$$f(y|x) = \frac{f(x, y)}{f(x)}$$

Bayes' Formula

$$\Rightarrow \underline{f(x, y) = f(y|x) \underline{f(x)}}$$

↓
Law of Total Probability

$$f(x, y) = \left(\frac{1}{x} \right) \left(\frac{\lambda^2 x e^{-\lambda x}}{\Gamma(2)} \right) I_{[0, x]}(y) I_{[0, \infty)}(x)$$



$$I_{[0, x]}(y) \cdot I_{[0, \infty)}(x)$$

$$= \{(x, y) \in \mathbb{R}^2 \mid 0 \leq x < \infty, 0 \leq y \leq x\}$$

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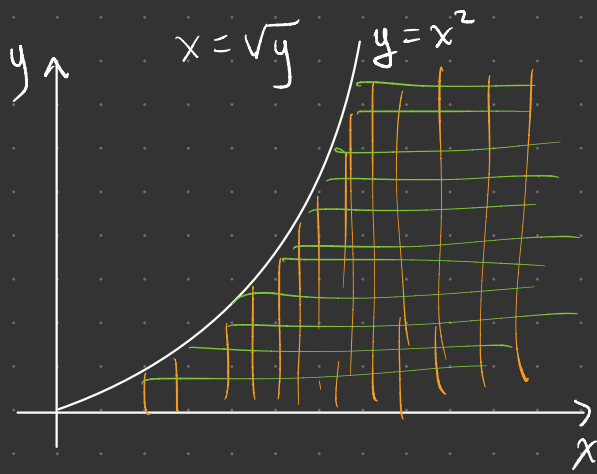
$$= I_{[0, \infty)}(y) \cdot I_{[y, \infty)}(x)$$

$$f(y) = \int_{-\infty}^{\infty} \left(\frac{1}{x} \right) \left(\frac{\lambda^2 x e^{-\lambda x}}{\Gamma(\lambda)} \right) \underbrace{I_{[0, x]}(y) I_{[0, \infty)}(x)}_{I_{[0, \infty)}(y) I_{[y, \infty)}(x)} dx$$

$$= \int_{-\infty}^{\infty} \left(\frac{1}{x} \right) \left(\frac{\lambda^2 x e^{-\lambda x}}{\Gamma(\lambda)} \right) I_{[0, \infty)}(y) I_{[y, \infty)}(x) dx$$

$$= \int_y^{\infty} \frac{1}{x} \left(\frac{\lambda^2 x e^{-\lambda x}}{\Gamma(\lambda)} \right) I_{[0, \infty)}(y) dx = G(\infty) - G(y)$$

G is the antiderivative of the integrand.



$$\{(x, y) \mid 0 \leq x < \infty, 0 \leq y \leq x^2\}$$

$$\{(y, x) \mid 0 \leq y < \infty, \sqrt{y} \leq x < \infty\}$$

$$P_r(A) = \mathbb{E}[1_A]$$

Remember! $\int_a^b f(x) dx = F(b) - F(a)$

where F is the antiderivative of f i.e., the indefinite integral

$$E(Y) = \int_{-\infty}^{\infty} y f_Y(y) dy$$

$$E(Y|X=x) = h(x)$$

$$= \int_{-\infty}^{\infty} E(Y|X=x) f_X(x) dx$$

Independence (Statistical) : $f(x, y) = f(x) f(y)$

If $f(x, y) \neq f(x) f(y)$ NOT Independent.

$$f(x, y)$$

$$E(X) = \int_{-\infty}^{\infty} x f_x(x) dx$$

$$\text{Cor}(X, Y) = E(YX) - E(X)E(Y)$$



$$E(Y) = \int_{-\infty}^{\infty} y f_y(y) dy$$

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy f(x, y) dx dy$$

$$\text{Var}(X) = E(X^2) - (E(X))^2$$

A new r.v.

$$Z = E(Y|X) = w(X)$$

$$\text{BLP } Y|X: E^*(Y|X) = \alpha + \beta X$$

$$\text{where } \alpha = E(Y) - \beta E(X)$$

$$\beta = \frac{\text{Cov}(X, Y)}{\text{Var}(X)}$$

$$\text{BLP } Z|X: E^*(Z|X) = \hat{\alpha} + \hat{\beta} X$$

$$\text{where } \hat{\alpha} = E(Z) - \hat{\beta} E(X)$$

$$\hat{\beta} = \frac{\text{Cov}(Z, X)}{\text{Var}(X)}$$

$$\frac{d}{dx} E[h(x)] = E\left[\frac{dh(x)}{dx}\right]$$

$$E(XYZ) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xyz f(x,y,z) dx dy dz$$